# STPM 2022 PHYSICS 960/1

## Questions and Answers

*All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.*

## Section A [15 marks]

**1.** Which is the scalar product of two vectors?

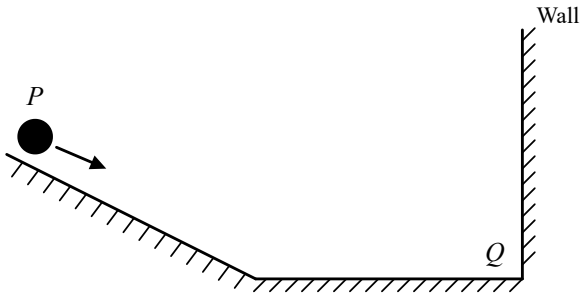
A Impulse B Torque C Work D Area

**Answer: C**

Work done is the scalar (dot) product of force and displacement: $W = \mathbf{F} \cdot \mathbf{s} = Fs \cos \theta$.

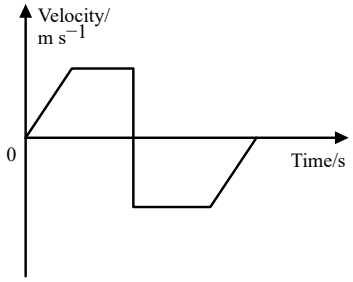
**Explanation:** A scalar product of two vectors gives a scalar. Impulse (force × time) is a vector, and torque and area are vector (cross) products of two vectors. Work has no direction, so it is the scalar product.

**2.** A ball is released from rest at *P* and the ball moves along a smooth slope and horizontal surface as shown in the diagram below.

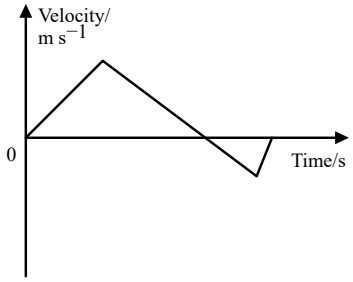



The ball rebounds elastically at *Q* and stops momentarily at *P*. Which velocity-time graph represents the motion of the ball?

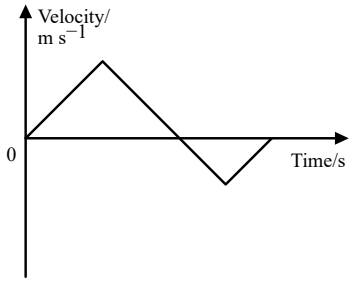
A



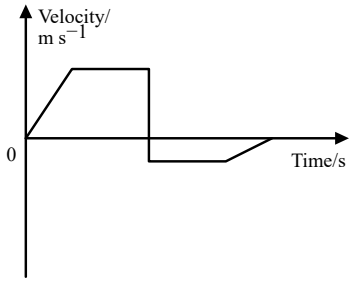

B



C




D




**Answer: A**

Down the smooth slope: the speed increases uniformly (straight line rising).

On the horizontal surface: constant speed.

Elastic rebound at $Q$: the velocity reverses with the same magnitude (constant negative value).

Up the slope: the speed decreases uniformly to zero at $P$.

**Explanation:** Velocity is a vector, so after the rebound it becomes negative. In an elastic collision no kinetic energy is lost, so the negative velocity has the same size as before, and the ball can climb back to $P$. Graph A is the only one with equal positive and negative constant sections.

**3.** An object of mass $2M$ moves with speed $v$ and collides head-on with another object of mass $M$ which moves with speed $\frac{1}{2}v$ in the opposite direction as shown in the diagram below.




If the collision is an inelastic collision, what is the speed of both objects after the collision?

A 0 B $\frac{1}{2}v$ C $\frac{5}{6}v$ D $\frac{\sqrt{3}}{2}v$

**Answer: B**

Taking the direction of $2M$ as positive, the objects stick together (common velocity):

$$(2M)v + M(-\tfrac{1}{2}v) = (3M)V$$

$$\tfrac{3}{2}Mv = 3MV \Longrightarrow V = \tfrac{1}{2}v$$

**Explanation:** Momentum is a vector, so the momentum of the mass moving the other way is negative. In a perfectly inelastic collision the objects move off together, so the total mass after the collision is $3M$.

**4.** A block of mass of 4.0 kg slides down a plane which is inclined at 35°. Then, a force of 16 N is applied parallel to the inclined plane to push the block up and moves at a constant speed. What is the coefficient of kinetic friction between the block and the surface of the inclined plane?

A 0.20 B 0.36 C 0.50 D 0.70

**Answer: A**

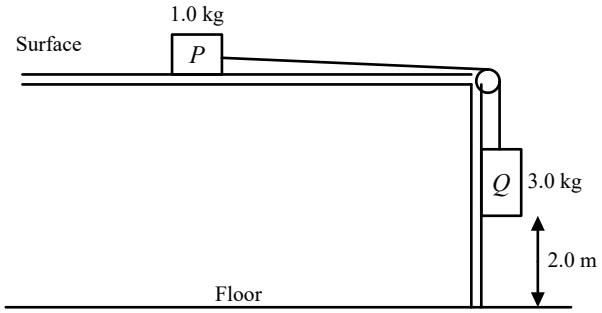
The block moves down the plane at constant speed, so friction acts up the plane together with the 16 N force:

$$mg \sin 35° = 16 + \mu mg \cos 35°$$

$$\mu = \frac{(4.0)(9.81)\sin 35° - 16}{(4.0)(9.81)\cos 35°} = \frac{6.51}{32.1} = 0.20$$

**Explanation:** At constant speed the resultant force is zero. The component of the weight down the slope (22.5 N) is balanced by the 16 N force and the kinetic friction, which always opposes the motion. (If the block were pushed up the slope, 16 N would be less than the weight component alone, so this reading is the only one that works.)

**5.** A block of mass of 1.0 kg, $P$, is placed on a smooth surfaced table and $P$ is connected to a block of mass 3.0 kg, $Q$, by a cord passing a frictionless pulley as shown in the diagram below.




If $Q$ is allowed to move downwards, what is the velocity of the block as the block touches the floor?

A $4.4\ \text{m s}^{-1}$ B $5.4\ \text{m s}^{-1}$ C $6.3\ \text{m s}^{-1}$ D $7.2\ \text{m s}^{-1}$

**Answer: B**

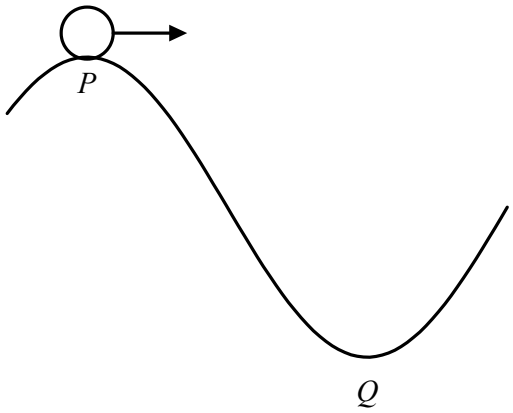
Loss of potential energy of $Q$ = gain of kinetic energy of $P$ and $Q$ (the table is smooth):

$$m_Q gh = \frac{1}{2}(m_P + m_Q)v^2 \Rightarrow (3.0)(9.81)(2.0) = \frac{1}{2}(4.0)v^2$$

$$v = 5.4\ \text{m s}^{-1}$$

**Explanation:** Both blocks move together with the same speed, so the kinetic energy involves the total mass. Alternatively, $a = \frac{m_Q g}{m_P + m_Q} = 7.36\ \text{m s}^{-2}$ and $v^2 = 2as$. Using only the mass of $Q$ gives option C.

**6.** A ball moves along a smooth curved track as shown in the diagram below.




If the velocities of the ball at $P$ and $Q$ are $V_P$ and $V_Q$ respectively, the net forces acted on the ball at $P$ and $Q$ are $R_P$ and $R_Q$ respectively. Which are the correct comparisons between $R_Q$ and $R_P$, and $V_Q$ and $V_P$?

A $R_Q > R_P$, $V_Q > V_P$
B $R_Q > R_P$, $V_Q < V_P$
C $R_Q < R_P$, $V_Q > V_P$
D $R_Q < R_P$, $V_Q < V_P$

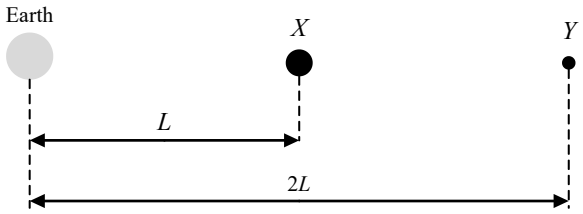
**Answer: A**

$Q$ is lower than $P$, so the ball has more kinetic energy there: $V_Q > V_P$.

Net (centripetal) force = $\frac{mv^2}{r}$, so the larger speed at $Q$ gives $R_Q > R_P$

**Explanation:** On a smooth track mechanical energy is conserved, so the ball is fastest at the lowest point. The net force on a ball moving along a curve points towards the centre of curvature and equals $\frac{mv^2}{r}$: at the crest it is $mg - N$, in the trough it is $N - mg$.

**7.** A star of mass $2M$, $X$, and a star of mass $M$, $Y$, are at a distance $L$ and $2L$ from the centre of the Earth respectively as shown in the diagram below.




What is the ratio of the force that acts on the Earth by $X$ to the force that acts on the Earth by $Y$?

A 1 : 2 B 1 : 4 C 4 : 1 D 8 : 1

**Answer: D**

$$F \propto \frac{M}{d^2}: \frac{F_X}{F_Y} = \frac{\frac{2M}{L^2}}{\frac{M}{(2L)^2}} = \frac{2}{\frac{1}{4}} = 8$$

Ratio 8 : 1

**Explanation:** The force depends on the mass of each star and on the inverse square of its distance from the Earth. $X$ is twice as massive and half as far away, so its force is $2 \times 2^2 = 8$ times larger.

**8.** The mass and radius of Mars are $6.46 \times 10^{23}$ kg and $3.39 \times 10^6$ m respectively. What is the escape velocity of Mars?

A $2.52 \text{ km s}^{-1}$ B $3.57 \text{ km s}^{-1}$ C $5.04 \text{ km s}^{-1}$ D $7.14 \text{ km s}^{-1}$

**Answer: C**

$$v = \sqrt{\left(\frac{2GM}{R}\right)} = \sqrt{\left(\frac{2(6.67 \times 10^{-11})(6.46 \times 10^{23})}{3.39 \times 10^6}\right)} = 5.04 \times 10^3 \text{ m s}^{-1} = 5.04 \text{ km s}^{-1}$$

**Explanation:** The escape velocity is found by equating the kinetic energy at launch to the energy needed to reach infinity, $\frac{1}{2}mv^2 = \frac{GMm}{R}$. Forgetting the factor 2 gives option B (3.57 km s$^{-1}$, the orbital speed at the surface).

**9.** The pressure of a fixed mass of an ideal gas from kinetic theory is given by $p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$ where standard symbols are used. What quantities are represented by $m$ and $N$?

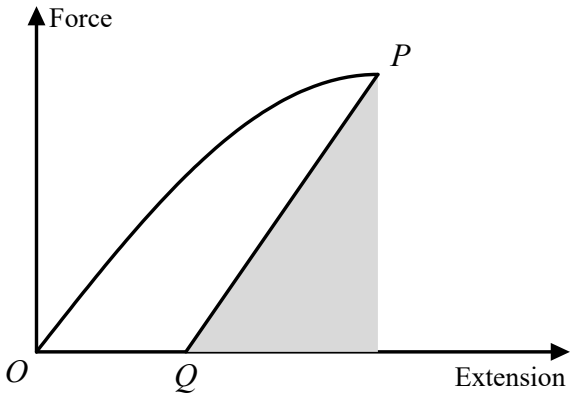
| | $m$ | $N$ |
|---|---|---|
| **A** | Mass of a molecule | Number of molecules |
| **B** | Mass of a molecule | Avogadro's number |
| **C** | Molar mass | Number of molecules |
| **D** | Molar mass | Avogadro's number |

**Answer: A**

In $p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$, $m$ is the mass of one molecule and $N$ is the number of molecules, so $Nm$ is the total mass of the gas.

**Explanation:** Since $Nm/V$ is the density of the gas, the equation can also be written $p = \frac{1}{3}\rho\langle c^2 \rangle$. Avogadro's number and molar mass refer to one mole, not to the whole sample.

**10.** The graph of force against extension of a wire is shown below.




The wire is stretched with increasing force just beyond the yield point, *P*. Then, the force is slowly decreased to zero at *Q* as shown in the graph above. Which statement is **not** true about the graph?

A The permanent extension of the wire is *OQ*.
B The area of the shaded region represents the strain energy in the wire.
C The area of *OPQ* represents the energy that can be recovered when the force is reduced to zero.
D The area under the graph *OP* represents the work done to stretch the wire just beyond the yield point.

**Answer: C**

When the force is reduced, the wire follows the line *PQ* and keeps a permanent extension *OQ* (A is true). The shaded area under *PQ* is the elastic strain energy that is released (B is true).

The area *OPQ* between the loading and unloading lines is the energy lost as heat in the plastic deformation; it cannot be recovered, so C is not true.

**Explanation:** The total work done in stretching the wire is the area under the loading curve *OP* (D is true). When the load is removed only the energy under the unloading line is returned; the rest, the area enclosed between the two lines, has been converted to heat.

**11.** The speed of six molecules of a gas are $u$, $2u$, $3u$, $3u$, $4u$, and $6u$ respectively. What is the root mean square speed of all the molecules?

A $2.96u$ B $3.00u$ C $3.17u$ D $3.54u$

**Answer: D**

$$c_{\text{rms}} = \sqrt{\left(\frac{u^2 + 4u^2 + 9u^2 + 9u^2 + 16u^2 + 36u^2}{6}\right)} = \sqrt{\left(\frac{75}{6}\right)}u = 3.54u$$

**Explanation:** Root mean square: square each speed, find the mean of the squares, then take the square root. It is always larger than the ordinary mean ($3.17u$, option C) because squaring gives more weight to the fast molecules.

**12.** The molar heat capacity of an ideal gas at constant pressure is $\frac{7}{2}R$. Which is true about the type and the degree of freedom of the gas at room temperature?

| | *Type of gas* | *Degree of freedom of the gas* |
|---|---|---|
| **A** | Monatomic | 3 |
| **B** | Diatomic | 5 |
| **C** | Diatomic | 7 |
| **D** | Polyatomic | 6 |

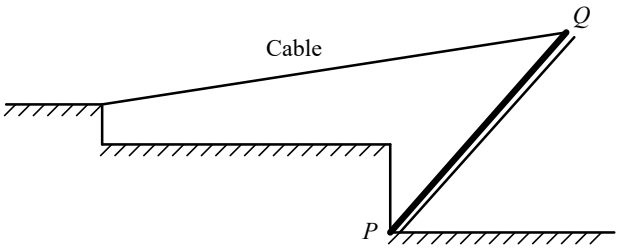
**Answer: B**

$$C_V = C_p - R = \frac{7}{2}R - R = \frac{5}{2}R = \frac{f}{2}R \Longrightarrow f = 5$$

Five degrees of freedom: a diatomic gas (3 translational + 2 rotational).

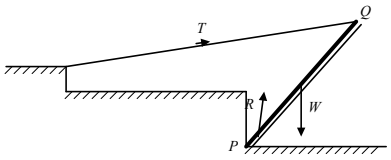
**Explanation:** For an ideal gas $C_V = \frac{f}{2}R$ and $C_p = C_V + R$. A monatomic gas has $C_p = \frac{5}{2}R$; a diatomic gas at room temperature has $C_p = \frac{7}{2}R$. Do not confuse $\frac{7}{2}R$ with 7 degrees of freedom.

**13.** A uniform rod, $PQ$, of weight $W$ is supported by a cable as shown in the diagram below.

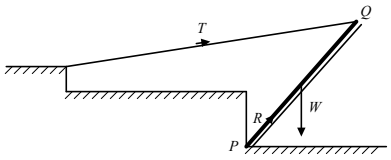



$R$ is the reaction force on the rod at $P$ and $T$ is the tension in the cable. Which diagram shows the correct forces that acts on the rod?

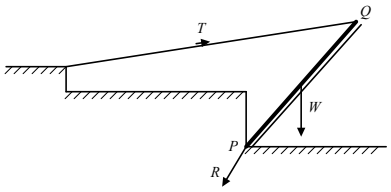
A



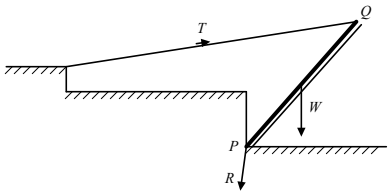

B




C




D



**Answer: A**

The rod is in equilibrium under three forces: $W$ (down, at the centre), $T$ (along the cable) and $R$ at $P$. Three non-parallel forces in equilibrium must pass through one point, so $R$ points from $P$ upwards towards the point where the lines of $T$ and $W$ meet, as in diagram A.

**Explanation:** The ground at $P$ pushes on the rod, so $R$ cannot point downwards into the ground (C and D are wrong). For the moments about any point to balance, the three lines of action must meet at a single point.

**14.** Which is **not** true about the movement of fluid in natural convection?

A Due to temperature gradient
B Due to pressure difference
C Due to mechanical means
D Due to density difference

**Answer: C**

Natural convection is caused by density differences (resulting from temperature differences and the pressure differences they produce), not by mechanical means. Movement caused by a fan or pump is forced convection.

**Explanation:** In natural convection warmer fluid expands, becomes less dense and is pushed upwards by the surrounding denser fluid (buoyancy). Any mechanical device that moves the fluid makes it forced convection.

**15.** A material has high specific heat capacity and thermal conductivity. Under the same conditions, which is true about the physical properties of the material?

A Easier to raise the temperature but the flow of heat is at lower rate
B Easier to raise the temperature but the flow of heat is at higher rate
C Harder to raise the temperature but the flow of heat is at lower rate
D Harder to raise the temperature but the flow of heat is at higher rate

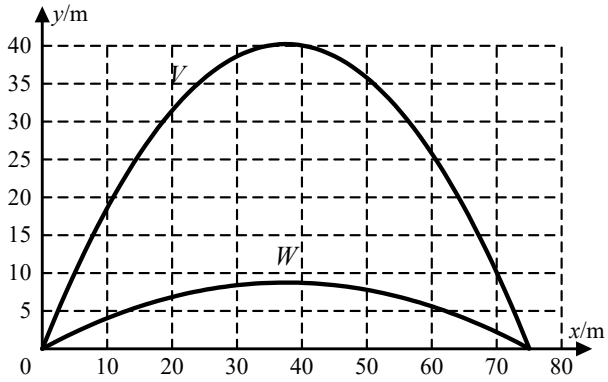
**Answer: D**

High specific heat capacity: a lot of heat is needed for each degree rise, so it is harder to raise the temperature.
High thermal conductivity: heat flows through it at a higher rate.

**Explanation:** Specific heat capacity describes how much energy is stored per kelvin; thermal conductivity describes how quickly heat travels through the material. They are independent properties, so a material can have both high.

## Section B [15 marks]

**16.** Two projectiles, $P$ and $Q$, are launched simultaneously into two trajectories, $V$ and $W$, respectively and both projectiles landed on the same point as shown in the graph below.




$P$ is launched at 65° from horizontal with the initial speed of 31 m s$^{-1}$. $Q$ is launched at 25° from horizontal.

[Assume that the air resistance is negligible.]

(a) Calculate the time of flight of $P$. [2]

**Answer**

$t = \frac{2u \sin\theta}{g} = \frac{2(31)\sin 65°}{9.81}$ 1

$t = 5.73$ s 1

*[Accept 5.7 s]*

**2**

**Explanation:** The projectile lands at the launch height, so the vertical displacement is zero: $0 = (u \sin\theta)t - \frac{1}{2}gt^2$ gives the time of flight.

(b) Determine the initial speed of $Q$. [4]

**Answer**

Range of $P$: $R = \frac{u^2 \sin 2\theta}{g} = \frac{31^2 \sin 130°}{9.81} = 75.0$ m 1

For $Q$: $R = \frac{u_Q^2 \sin 50°}{9.81}$ 1

Both land at the same point: $\frac{u_Q^2 \sin 50°}{9.81} = 75.0$ 1

$u_Q = 31$ m s$^{-1}$ 1

**4**

**Explanation:** Launch angles of 65° and 25° add up to 90°, and sin 130° = sin 50°, so complementary angles give the same range for the same launch speed. Hence $Q$ must also be launched at 31 m $s^{-1}$, even though its flight is much shorter in time.

(c) Which trajectory is most affected by the presence of air resistance? Give your reason. [2]

**Answer**

| | |
|---|---|
| Trajectory $V$ (projectile $P$) | 1 |
| It has the longer time of flight / travels a longer path through the air, so air resistance acts on it for longer | 1 |
| | **2** |
| *Total for question* | **8** |

**Explanation:** The two projectiles do not have the same time of flight: $P$ stays in the air for 5.73 s but $Q$ for only about 2.7 s. The longer the projectile is in the air, the more its motion is affected by air resistance.

**17.** (a) A copper rod of length 45.0 cm has a cross-sectional area of 1.25 $cm^2$. One end of the copper is heated to 100 °C and the other end is placed in an ice bath of 0 °C. When the copper rod reaches its steady state, calculate

(i) the temperature gradient along the rod [2]

**Answer**

| | |
|---|---|
| Temperature gradient = $\frac{\Delta\theta}{\Delta x} = \frac{100 - 0}{0.450}$ | 1 |
| = 222 K $m^{-1}$ (°C $m^{-1}$) | 1 |
| *[A negative sign, −222 K $m^{-1}$, is also accepted]* | |
| | **2** |

**Explanation:** In the steady state the temperature falls uniformly along a well-lagged rod, so the gradient is the total temperature difference divided by the length (in metres).

(ii) the rate of heat flow in the rod

[Thermal conductivity for copper is 385 W $m^{-1}$ $K^{-1}$.] [2]

**Answer**

| | |
|---|---|
| $\frac{Q}{t} = kA\frac{\Delta\theta}{\Delta x} = (385)(1.25 \times 10^{-4})(222)$ | 1 |
| = 10.7 W | 1 |
| | **2** |

**Explanation:** Convert the area from $cm^2$ to $m^2$ (1.25 $cm^2$ = 1.25 × $10^{-4}$ $m^2$). The rate of heat flow is a positive quantity.

(b) Explain the mechanism of heat conduction through the copper rod. [3]

**Answer**

| | |
|---|---|
| Atoms at the hot end vibrate with larger amplitude and pass energy to neighbouring atoms through collisions (lattice vibration) | 1 |
| Copper has free electrons, which gain kinetic energy at the hot end | 1 |
| The free electrons diffuse quickly to the cold end and transfer energy by colliding with the atoms there | 1 |
| | **3** |
| *Total for question* | **7** |

**Explanation:** Use the words atoms and free electrons (not 'particles' or 'molecules'). Metals such as copper conduct well mainly because their free electrons carry energy quickly along the rod.

## Section C [30 marks]

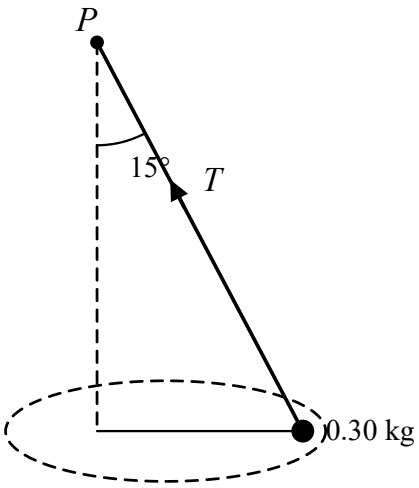
**18.** (a) A small sphere of mass $m$ is attached to an end of a string of length $R$. The sphere is whirled in vertical circular path about a fixed point, $O$, with constant speed $v$. Derive an expression for the tension in the string at any point which makes an angle $\theta$ with the vertical when passing through $O$. [4]

**Answer**

| | |
|---|---|
| At angle $\theta$ from the vertical (measured from the lowest point), the forces are the tension $T$ towards $O$ and the weight $mg$; the component of the weight along the string is $mg\cos\theta$ away from $O$ | 1 |
| Resultant force towards the centre = $T - mg\cos\theta$ | 1 |
| $T - mg\cos\theta = \frac{mv^2}{R}$ | 1 |
| $T = m(\frac{v^2}{R} + g\cos\theta)$ | 1 |
| | **4** |

**Explanation:** Resolve the weight along and perpendicular to the string. Only the forces along the string give the centripetal force. Check: at the bottom ($\theta = 0$) $T = \frac{mv^2}{R} + mg$, the largest value; at the top ($\theta = 180°$) $T = \frac{mv^2}{R} - mg$.

(b) A ball of mass 0.30 kg is attached to a string of length 1.5 m and suspended from $P$ to make a conical pendulum as shown in the diagram below.




The ball whirled around in a horizontal circle and the string makes an angle of 15° to the vertical axis.

(i) Determine the speed of the ball. [4]

**Answer**

Radius of the circle: $r = 1.5 \sin 15° = 0.388$ m — 1

Vertical: $T \cos 15° = mg$ — 1

Horizontal: $T \sin 15° = \frac{mv^2}{r}$ — 1

$\tan 15° = \frac{v^2}{rg} \Longrightarrow v = \sqrt{((0.388)(9.81) \tan 15°)} = 1.01 \text{ m s}^{-1}$ — 1

**4**

**Explanation:** The radius of the horizontal circle is not the length of the string: it is the horizontal distance of the ball from the vertical axis, 1.5 sin 15°. Dividing the two force equations removes the tension.

(ii) Determine the centripetal acceleration of the ball. [2]

**Answer**

$a = \frac{v^2}{r} = \frac{1.01^2}{0.388}$ — 1

$a = 2.63 \text{ m s}^{-2}$ $(= g \tan 15°)$ — 1

**2**

**Explanation:** The centripetal acceleration of a conical pendulum is $g \tan \theta$, which is a quick check on the answer.

(iii) Calculate the tension in the string to keep the ball in circular motion. [2]

**Answer**

$T = \frac{mg}{\cos 15°} = \frac{(0.30)(9.81)}{\cos 15°}$ 1

$T = 3.05$ N 1

**2**

**Explanation:** The vertical component of the tension balances the weight, so the tension is slightly larger than the weight (2.94 N).

(c) State **three** characteristics of the centripetal acceleration for a uniform circular motion of the ball. [3]

**Answer**

• Its magnitude is constant, $a = \frac{v^2}{r} = r\omega^2$ 1

• It is always directed towards the centre of the circle 1

• It is perpendicular to the velocity (its direction changes continuously) 1

**3**

*Total for question* **15**

**Explanation:** Describe the acceleration itself: its size, its direction and how the direction changes. Saying only what provides the centripetal force (e.g. the tension) does not answer the question.

**19.** (a) What is meant by *the centre of gravity* of an object? [1]

**Answer**

The point at which the whole weight of the object appears to act. 1

**1**

**Explanation:** Use the word weight, not mass: the point where the mass appears to be concentrated is the centre of mass.

(b) State **two** conditions for an equilibrium of a rigid body. [2]

**Answer**

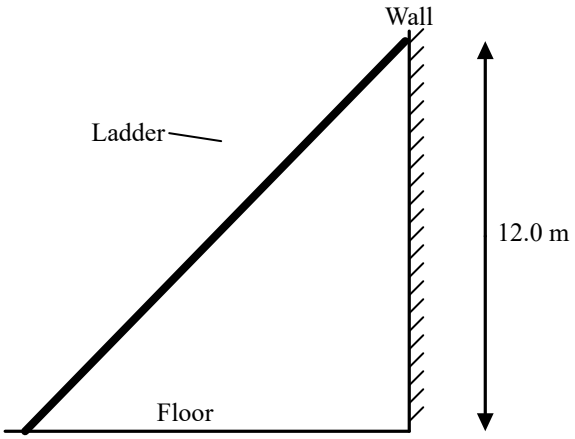
The resultant force on the body is zero ($\Sigma F = 0$) 1

The resultant torque (moment) about any point is zero ($\Sigma \tau = 0$) 1

**2**

**Explanation:** The first condition prevents translational acceleration and the second prevents rotational acceleration.

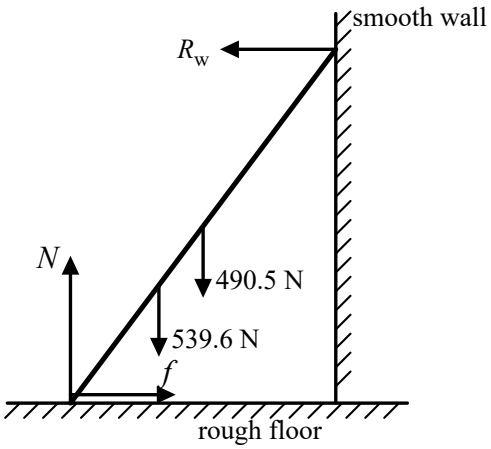
(c) A ladder of length 15.0 m and mass 55.0 kg rests against a smooth wall with the top end is 12.0 m from the floor as shown in the diagram below.




The centre of gravity of the ladder is located at one third of the length from the bottom end. A boy of mass 50.0 kg climbs up the ladder until centre of gravity of the boy is half of the length of the ladder.

(i) Sketch all the forces which act on the ladder. [3]

**Answer**




| | |
|---|---|
| Weights of the ladder (539.6 N, at one third of the length) and of the boy (490.5 N, at half the length) | 1 |
| Normal reaction of the smooth wall, horizontal | 1 |
| Normal reaction and friction (towards the wall) at the floor | 1 |
| | **3** |

**Explanation:** A smooth wall can only push perpendicular to itself, so its force is horizontal. The floor provides an upward normal reaction and friction; friction acts towards the wall because the foot of the ladder tends to slide away from it. The boy's weight acts on the ladder at his position.

(ii) Determine the forces exerted by the wall and the floor on the ladder. [6]

**Answer**

The foot of the ladder is $\sqrt{(15.0^2 - 12.0^2)} = 9.0$ m from the wall; horizontal distances: ladder's weight 3.0 m, boy 4.5 m — 1

Moments about the foot: $R_w(12.0) = (55.0)(9.81)(3.0) + (50.0)(9.81)(4.5)$ — 1

$R_w = 319$ N (force exerted by the wall) — 1

Vertical: $N = (55.0 + 50.0)(9.81) = 1030$ N — 1

Horizontal: $f = R_w = 319$ N — 1

Resultant force of the floor = $\sqrt{(1030^2 + 319^2)} = 1.08 \times 10^3$ N at $\tan^{-1}\frac{1030}{319} = 72.8°$ to the floor — 1

**6**

**Explanation:** Take moments about the foot of the ladder so that both unknown floor forces drop out. Each weight acts vertically, so its moment arm is a horizontal distance: one third and one half of the 9.0 m horizontal span.

(iii) If the coefficient of static friction between the ladder and the floor is 0.45, what is the length of the ladder that could be climbed by the boy before the ladder slips? [3]

**Answer**

Maximum friction: $f = \mu N = (0.45)(1030) = 464$ N $= R_w$ — 1

Moments about the foot: $(464)(12.0) = (55.0)(9.81)(3.0) + (50.0)(9.81)x \Longrightarrow x = 8.04$ m (horizontal) — 1

Length along the ladder = $\frac{8.04}{\cos\theta} = \frac{8.04}{0.60} = 13.4$ m — 1

**3**

*Total for question* **15**

**Explanation:** The ladder is about to slip when friction reaches its maximum value $\mu N$. The normal reaction $N$ does not change as the boy climbs (it always equals the total weight), but the wall reaction, and therefore the friction needed, increases as he goes higher.

**20.** (a)

(i) What is meant by *molar heat capacity at constant pressure* of an ideal gas? [1]

**Answer**

The quantity of heat required to raise the temperature of 1 mole of the gas by 1 K at constant pressure. — 1

**1**

**Explanation:** Molar heat capacity is per mole and per kelvin; at constant pressure the gas also expands and does work, so $C_p$ is larger than $C_V$.

(ii) A hydrogen gas of 1.00 mol is heated at constant pressure from 300 K to 420 K. Determine the work done by the gas.
[Molar heat capacity of hydrogen at constant pressure and at constant volume are 28.8 J mol$^{-1}$ K$^{-1}$ and 20.4 J mol$^{-1}$ K$^{-1}$ respectively.] [3]

**Answer**

| | |
|---|---|
| $Q = nC_p\Delta T$ and $\Delta U = nC_V\Delta T$ | 1 |
| $W = Q - \Delta U = n(C_p - C_V)\Delta T = (1.00)(28.8 - 20.4)(420 - 300)$ | 1 |
| $W = 1008$ J | 1 |
| | **3** |

**Explanation:** By the first law, the work done by the gas at constant pressure is the difference between the heat supplied and the increase in internal energy. Note that $C_p - C_V = 8.4$ J mol$^{-1}$ K$^{-1}$, close to $R$.

(b) A perfectly insulated cylinder with a piston is filled with an ideal gas. The piston is pushed in with constant pressure to compress the gas. By using thermodynamic process, explain the changes of the temperature in the system. [4]

**Answer**

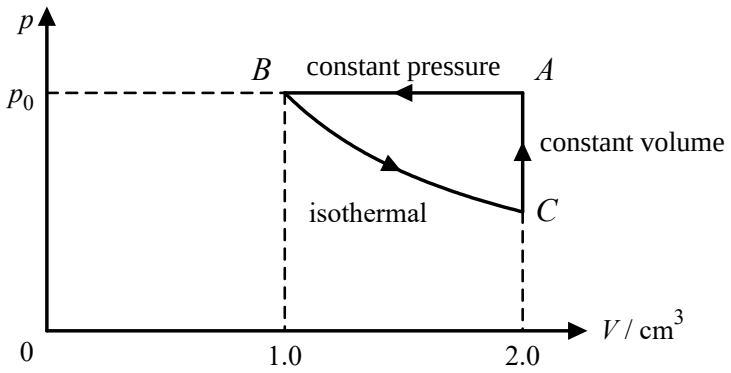
| | |
|---|---|
| The cylinder is perfectly insulated, so the process is adiabatic: $Q = 0$ | 1 |
| Work is done on the gas as it is compressed ($W$ is negative) | 1 |
| From $Q = \Delta U + W$: $\Delta U = -W > 0$, the internal energy increases | 1 |
| So the temperature of the gas increases | 1 |
| | **4** |

**Explanation:** No heat can leave an insulated cylinder, so all the work done on the gas by the piston goes into its internal energy. The internal energy of an ideal gas depends only on temperature, so the gas gets hotter (as in a bicycle pump).

(c) An ideal gas of 2.0 cm$^3$ at atmospheric pressure is cooled at constant pressure to 1.0 cm$^3$. The gas expands isothermally back to 2.0 cm$^3$. Then the pressure of the gas is increased at constant

volume until the original pressure is reached. Sketch and label a *p*-*V* graph of the thermodynamic processes. [2]

**Answer**




| | |
|---|---|
| Correct shapes: horizontal line (constant pressure), curve (isothermal) and vertical line (constant volume) | 1 |
| Processes in the correct order and direction (arrows), returning to the original state | 1 |
| | **2** |

**Explanation:** Constant pressure gives a horizontal line, constant temperature gives a curve $pV$ = constant, and constant volume gives a vertical line. Cooling at constant pressure reduces the volume, so the first line goes to the left.

(d) An insulated cylinder fitted with a frictionless light piston at 300 K contains 2.00 mol of ideal gas of helium. The gas expands isothermally such that the pressure changes from $1.22 \times 10^5$ Pa to $0.405 \times 10^5$ Pa.

(i) Calculate the final volume of the gas. [3]

**Answer**

| | |
|---|---|
| $pV = nRT \Longrightarrow V_2 = \frac{nRT}{p_2}$ | 1 |
| $V_2 = \frac{(2.00)(8.31)(300)}{0.405 \times 10^5}$ | 1 |
| $V_2 = 0.123\ \text{m}^3$ | 1 |
| | **3** |

**Explanation:** The temperature stays at 300 K, so the final volume follows directly from the ideal gas equation with the final pressure (or from $p_1V_1 = p_2V_2$).

(ii) Calculate the work done by the gas. [2]

**Answer**

$W = nRT \ln \frac{p_1}{p_2} = (2.00)(8.31)(300) \ln \frac{1.22}{0.405}$ 1

$W = 5.50 \times 10^3$ J 1

**2**

*Total for question* **15**

**Explanation:** For an isothermal process $W = nRT \ln \frac{V_2}{V_1} = nRT \ln \frac{p_1}{p_2}$. Do not use the adiabatic formula: although the cylinder is insulated, the question states the expansion is isothermal.